COMPUTER VISION

Assignment 3 — Complete Solutions

B.Tech Level • Questions + Answers • Equations • Numerical Example • Diagrams

Topics covered:
Pinhole camera and perspective projection • Depth estimation • Binocular stereopsis • RANSAC and DLT • Epipolar geometry • Homography • Stereo rectification • Epipolar constraints • Stereo baseline trade-offs

This PDF includes the assignment questions followed by their corresponding B.Tech-level solutions.

Computer Vision — Assignment 3

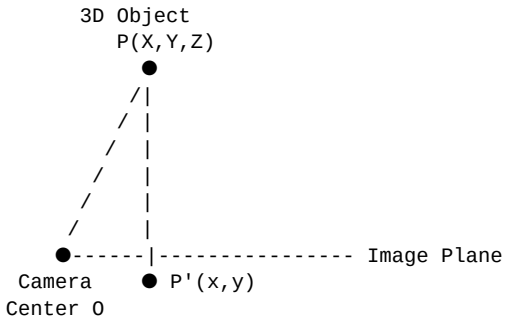
I'll solve each question at a **B.Tech level**, with the important equations and diagrams. The assignment focuses mainly on **camera geometry, stereo vision, epipolar geometry, homography, and stereo matching**.

Q1. Explain the pinhole-camera model and perspective projection. Explain why objects appear smaller when their distance from the camera increases.

Answer

The **pinhole camera model** is a simple mathematical model that describes how a 3D point in the real world is projected onto a 2D image plane.

Pinhole Camera Model



Let:

- (P(X,Y,Z)) = 3D point
- (P'(x,y)) = projected image point
- (f) = focal length
- (Z) = distance/depth of the point from camera

Using similar triangles:

$$(x)/(f)=(X)/(Z)$$

Therefore,

$$x=f(X)/(Z)$$

Similarly,

$$y=f(Y)/(Z)$$

These are the **perspective projection equations**.

Why do distant objects appear smaller?

From:

$$x=f(X)/(Z)$$

and

$$y=f(Y)/(Z)$$

we can see that image size is inversely proportional to depth:

$$\text{Image size} \propto (1)/(Z)$$

Therefore, as the object moves farther from the camera:

$$Zuparrow quadRightarrowquad x,ydownarrow$$

Example

Suppose:

$$f=50mm,quad X=100mm$$

For:

$$Z=1000mm$$

$$x=50(100)/(1000)=5mm$$

If the object moves to:

$$Z=2000mm$$

then:

$$x=50(100)/(2000)=2.5mm$$

So the projected size becomes **half**.

Conclusion

The pinhole model demonstrates the basic principle of perspective: **objects farther from the camera occupy a smaller portion of the image**.

Q2. What are the types of depth estimation? What is the ambiguity in single image view?

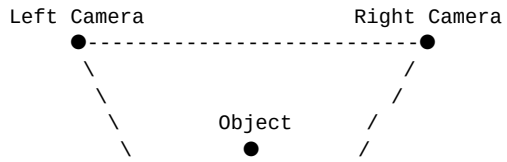
Answer

Depth estimation is the process of determining the distance of objects or scene points from the camera.

Major Types of Depth Estimation

1. Stereo Depth Estimation

Uses two cameras viewing the same scene from different positions.



The difference in the position of an object between the two images is called **disparity**.

Depth is calculated using:

$$Z=(fB)/(d)$$

where:

- (Z) = depth
- (f) = focal length
- (B) = baseline between cameras
- (d) = disparity

2. Monocular Depth Estimation

Uses only **one image**.

Depth is estimated from visual cues such as:

- Object size
- Perspective
- Shadows
- Texture
- Occlusion
- Familiar object dimensions

Modern computer vision systems can also use deep learning for monocular depth estimation.

3. Structure from Motion

Uses multiple images captured from a moving camera.

The movement of image features across frames is used to estimate:

- Camera motion
- 3D structure
- Depth

4. Depth from Focus/Defocus

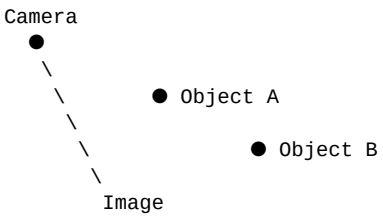
Depth is estimated using the amount of blur or focus of objects.

Objects at different distances may have different levels of sharpness.

Ambiguity in a Single Image

A single 2D image does not directly provide the actual 3D depth of every point.

For example:



Two objects of different sizes and distances can produce the **same image size**.

From perspective projection:

```
x=f(X)/(Z)
```

A change in (X) can compensate for a change in (Z).

Therefore, many different 3D scenes can produce the same 2D image.

This is called the **depth ambiguity** or **inverse projection ambiguity**.

Example

A small object close to the camera may appear the same size as a large object far away.

Hence:

```
Single image notRightarrow unique 3D depth
```

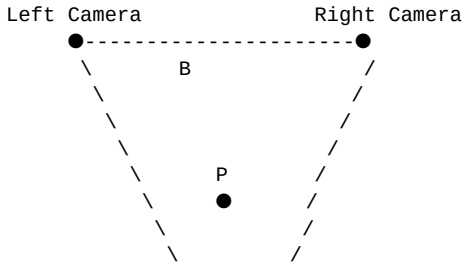
Additional information such as stereo images, motion, focus, or learned scene priors is needed.

Q3. Explain binocular stereopsis. A stereo-camera system has (f=800) pixels, (B=0.12m). For a scene point, disparity is (d=24) pixels. Calculate the depth.

Answer

Binocular stereopsis is the process of estimating the depth of a scene using two cameras, similar to how human eyes perceive depth.

The two cameras are separated by a distance called the **baseline (B)**.



The same object appears at different horizontal positions in the left and right images.

The difference is called **disparity**:

```
d=x_L-x_R
```

The stereo depth equation is:

```
Z=(fB)/(d)
```

Given:

```
f=800 pixels
```

```
B=0.12m
```

```
d=24 pixels
```

Substitute:

```
Z=(800 × 0.12)/(24)
```

```
Z=(96)/(24)
```

```
Z=4m
```

Final Answer

```
Depth=4 metres
```

Important Relationship

```
Z ∝ (1)/(d)
```

Therefore:

- Large disparity → object is closer
- Small disparity → object is farther away

Q3(b). Differentiate between RANSAC and DLT.

Answer

RANSAC and **DLT (Direct Linear Transform)** are commonly used in computer vision for estimating geometric models.

DLT

DLT estimates transformation parameters from corresponding points.

For example, for a homography:

```
x'=H x
```

where (H) is the transformation matrix.

DLT generally assumes that the input correspondences are reasonably correct.

RANSAC

RANSAC (**Random Sample Consensus**) is a robust estimation method that can handle **outliers**.

Basic procedure:

```
Randomly select minimum points
          ↓
Estimate model
          ↓
Count inliers
          ↓
Repeat many times
          ↓
Select model with most inliers
```

Comparison

FeatureDLTRANSAC		
Full form	Direct Linear Transform	Random Sample Consensus
Main purpose	Estimate transformation	Robust model estimation
Handles outliers	Poorly	Very well
Approach	Uses point correspondences	Randomly samples subsets
Speed	Usually faster	More computationally expensive
Typical use	Homography/camera estimation	Homography estimation with noisy matches

Key Point

DLT answers:

> **"What transformation fits these points?"**

RANSAC answers:

> **"What transformation fits the largest set of reliable points while ignoring outliers?"**

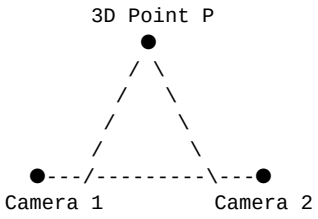
They can also be used **together**: RANSAC identifies reliable correspondences, and DLT estimates the transformation from those inliers.

Q4. Define epipole, epipolar plane, epipolar line and corresponding points.

Answer

These concepts are fundamental to **stereo vision**.

Consider two cameras observing the same 3D point (P).



1. Epipole

The **epipole** is the point where the line joining the two camera centers intersects the image plane of the other camera.

If the two camera centers are (C_1) and (C_2):

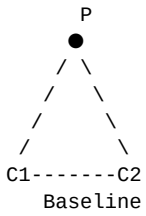
- (C_2) projected into camera 1 $\rightarrow$ epipole (e_1)
- (C_1) projected into camera 2 $\rightarrow$ epipole (e_2)

2. Epipolar Plane

The plane containing:

- Camera center (C_1)
- Camera center (C_2)
- 3D point (P)

is called the **epipolar plane**.



3. Epipolar Line

The intersection of the epipolar plane with an image plane is called the **epipolar line**.

Thus, the projection of a 3D point in one camera must lie on a corresponding epipolar line in the other camera.

4. Corresponding Points

The projections of the **same 3D point** in the two images are called corresponding points.

If:

```
Parrow p_L
```

in the left image and:

```
Parrow p_R
```

in the right image, then (p_L) and (p_R) are corresponding points.

Importance

Epipolar geometry reduces stereo matching from searching an entire 2D image to searching along a **1D epipolar line**.

Q5. Explain Homography. What are its applications?

Answer

A **homography** is a projective transformation that maps points from one image plane to another.

It is represented using a (3\times3) matrix (H).

In homogeneous coordinates:

```
x'=Hx
```

where:

```
H= [ h_11&h_12&h_13 h_21&h_22&h_23 h_31&h_32&h_33 ]
```

For a point:

```
x= [ xy1 ]
```

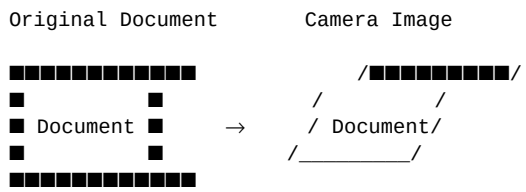
we obtain:

```
x'=Hx
```

and then convert the result back from homogeneous coordinates.

Example

Suppose an image of a rectangular document is captured at an angle.



A homography can transform the perspective-distorted document into a frontal view.

Applications

1. **Document rectification**
2. **Image stitching/panoramas**
3. **Perspective correction**
4. **Augmented reality**
5. **Image registration**
6. **Mapping images onto planes**
7. **Bird's-eye-view generation**

Important Point

Homography works especially well when:

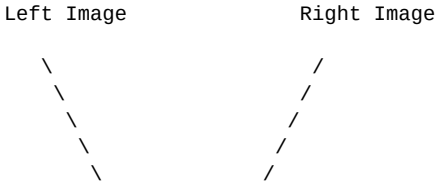
- the scene is planar, or
- the camera undergoes certain rotations/projective changes.

Q6. Define stereo-image rectification. Why is it required before stereo matching? Explain how rectification transforms corresponding epipolar lines into horizontal scan lines and reduces computational complexity.

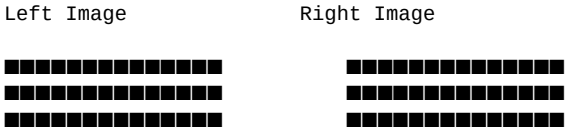
Answer

Stereo-image rectification is the process of transforming two stereo images so that their corresponding epipolar lines become **horizontal and aligned**.

Before rectification:



After rectification:



Why is Rectification Required?

Without rectification, a point in the left image could correspond to a point anywhere along a slanted epipolar line in the right image.

This makes matching more difficult.

After rectification:

```
y_L=y_R
```

Therefore, corresponding points lie on the **same horizontal scan line**.

The search changes from:

```
2D search
```

to:

```
1D horizontal search
```

Computational Advantage

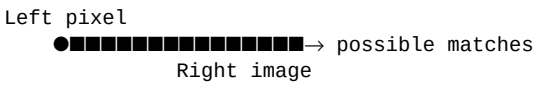
Suppose an image has:

```
W × H
```

pixels.

Without rectification, matching requires searching across two dimensions.

With rectification, for each pixel we mainly search along its horizontal row.



This significantly reduces the search space.

Benefits

- Faster stereo matching
- Easier correspondence search
- Simpler disparity calculation
- Better implementation using scan lines
- Improved computational efficiency

Q7. A point visible in the left image can appear anywhere in the right image before applying geometric constraints. Explain how epipolar geometry reduces this search from a two-dimensional region to a one-dimensional epipolar line. Why does this reduction improve both speed and matching accuracy?

Answer

Suppose a point (P) appears at:

```
p_L
```

in the left image.

Without any geometric constraint, its corresponding point in the right image could theoretically be anywhere:

```
Right Image

■■■■■■■■■■■■■■■■■■■■■■■■
■     ●             ■
■                   ■
■         ●         ■
■                   ■
■   ●               ■
■             ●     ■
■■■■■■■■■■■■■■■■■■■■■■■■
```

This requires a **2D search**.

Epipolar Constraint

Epipolar geometry states that the corresponding point must lie on a particular **epipolar line**.

Therefore:

```
Right Image

■■■■■■■■■■■■■■■■■■■■■■■■
■                   ■
■                   ■
■■■■■■■■■■■■●■■■■■■■■■■ ← Epipolar line
■                   ■
■                   ■
■■■■■■■■■■■■■■■■■■■■■■■■
```

Instead of searching the entire image, we search only along this line.

Mathematically, the epipolar constraint can be expressed using the fundamental matrix:

```
x_R^T F x_L=0
```

where:

- (F) = fundamental matrix
- (x_L) = point in left image
- (x_R) = corresponding point in right image

The equation defines the epipolar line on which (x_R) must lie.

Why does this improve speed?

Without epipolar geometry:

```
Search areaapprox W × H
```

With epipolar geometry:

```
Search areaapprox W
```

for a particular scan line.

Thus, the search becomes approximately:

```
2Darrow1D
```

This drastically reduces the number of candidate matches.

Why does it improve accuracy?

Searching the whole image may result in false matches because many pixels can have similar appearance.

Epipolar geometry eliminates impossible candidates.

Therefore:

```
Fewer candidatesarrowfewer false matches
```

So epipolar constraints improve both:

- **Computational speed**
- **Correspondence accuracy**

Q8. Analyze how baseline affects disparity, depth accuracy, occlusion and correspondence difficulty. Which system would be better for nearby objects, and which would be more suitable for distant objects?

The assignment gives two stereo-camera systems:

- **System 1:** Small baseline
- **System 2:** Large baseline

Both have the same focal length.

Answer

The fundamental stereo equation is:

```
Z=(fB)/(d)
```

where:

- (Z) = depth
- (f) = focal length

- (B) = baseline
- (d) = disparity

Rearranging:

```
d=(fB)/(Z)
```

Therefore, for the same object and focal length:

```
BuparrowRightarrow duparrow
```

So a larger baseline produces greater disparity.

1. Effect on Depth Accuracy

Depth is:

```
Z=(fB)/(d)
```

For a disparity error (\Delta d), approximately:

```
Delta Zapprox (Z^2)/(fB)Delta d
```

Therefore, increasing (B) reduces the effect of disparity error on depth estimation.

So:

```
Large baselinearrowbetter depth accuracy
```

especially for distant objects.

2. Effect on Disparity

From:

```
d=(fB)/(Z)
```

A larger baseline gives larger disparity.

Small baseline

```
BdownarrowRightarrow ddownarrow
```

Large baseline

```
BuparrowRightarrow duparrow
```

Larger disparity is easier to measure accurately in pixels.

3. Effect on Occlusion

A large baseline means the two cameras view the scene from significantly different positions.

Therefore, some objects visible in one camera may be hidden in the other camera.

```
Small baseline:

Camera 1 ●--● Camera 2
              \
               Object

Less viewpoint difference
→ Less occlusion
```

With a large baseline:

```
Camera 1 ●----------● Camera 2
             Object
```

```
Large viewpoint difference
→ More occlusion
```

Therefore:

```
Large baselinearrowmore occlusion
```

4. Effect on Correspondence Difficulty

A large baseline creates a larger viewpoint difference.

This causes:

- More occlusions
- Greater appearance differences
- Larger disparity search range

Therefore, correspondence can become more difficult.

A small baseline gives:

- Similar viewpoints
- Fewer occlusions
- Easier correspondence

but produces smaller disparity.

Small vs Large Baseline

PropertySmall BaselineLarge Baseline		
Disparity	Small	Large
Depth accuracy	Lower for distant objects	Higher
Occlusion	Less	More
Viewpoint difference	Small	Large
Correspondence	Easier	More difficult
Nearby objects	Better	Can be problematic
Distant objects	Less accurate	Better

Which System for Nearby Objects?

Small baseline is better for nearby objects.

Why?

Nearby objects can produce very large disparities when the baseline is large. This can cause:

- Large correspondence search range
- More occlusion
- Matching difficulties

A smaller baseline reduces these problems.

```
Nearby objectsarrowSmall baseline
```

Which System for Distant Objects?

Large baseline is better for distant objects.

For distant objects:

```
Z is large
```

and therefore:

```
d=(fB)/(Z)
```

can become very small.

With a small baseline, the disparity may become only a fraction of a pixel, making accurate depth estimation difficult.

Increasing (B) increases disparity:

```
BuparrowRightarrow duparrow
```

making depth estimation more precise.

```
Distant objectsarrowLarge baseline
```

Final Conclusion

```
Nearby objects &arrow Small baseline Distant objects &arrow Large baseline
```

The choice is therefore a trade-off between **disparity/depth accuracy** and **occlusion/correspondence difficulty**.